Fixation time and Probability of Cancer Cell in a Complete Graph

Prepared by : Abolfazl Khoshyari

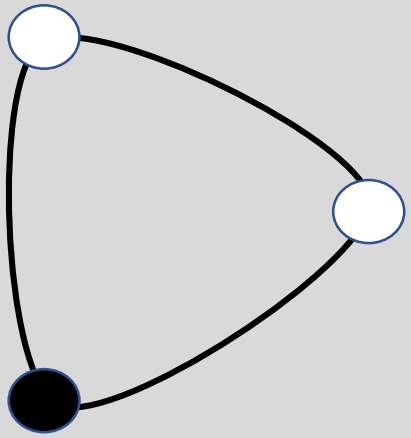
Supervisor : Moladad Nikbakht

University : Zanjan

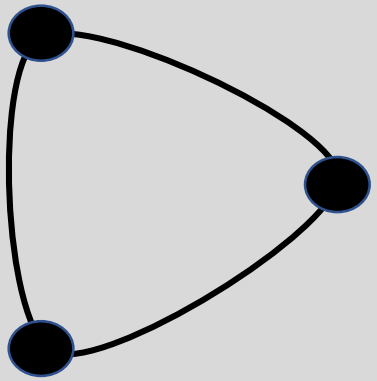
Date : 2024/12/29 ; (1403/10/9)

- **Objective**: study of cell progression

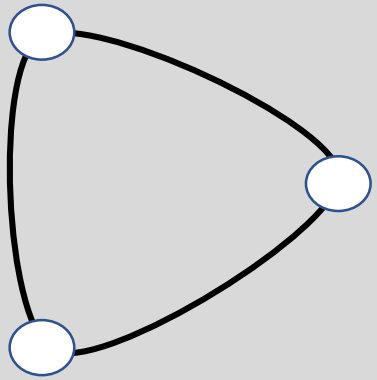
- Graph as Tissue:



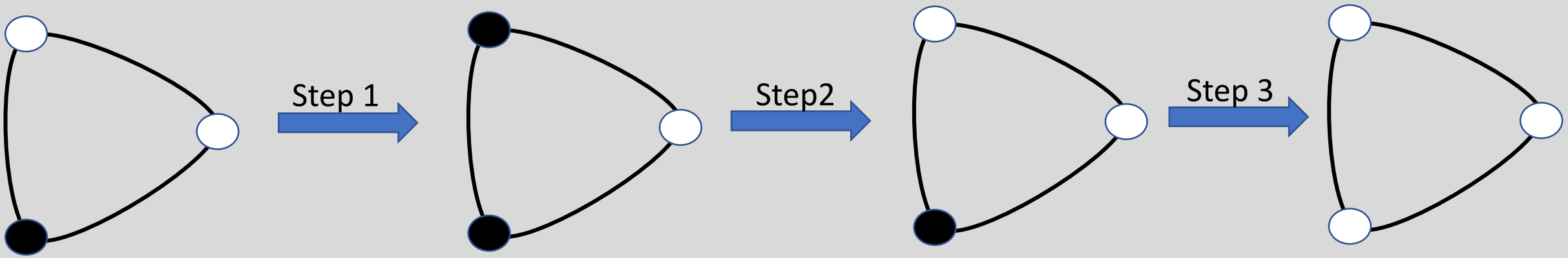
- Final state:

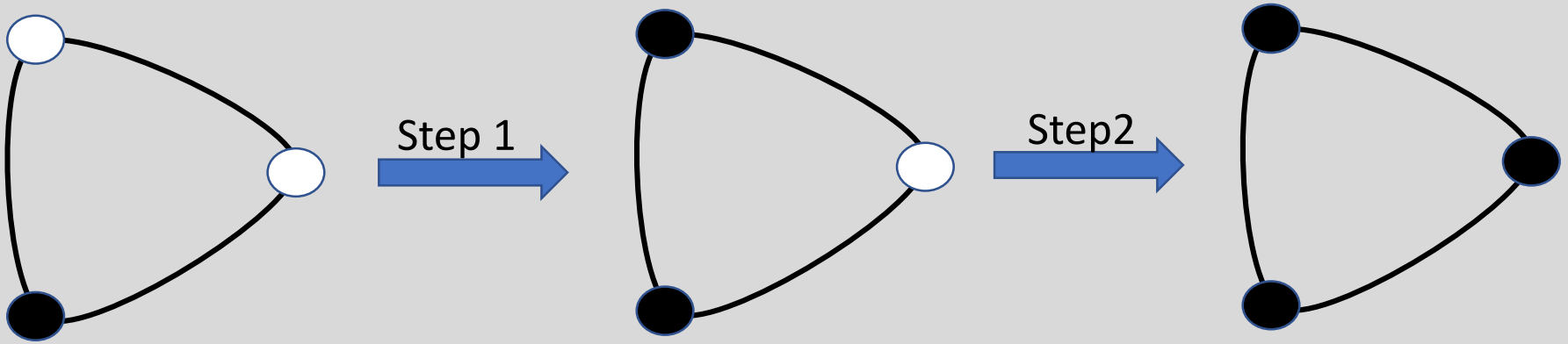


or



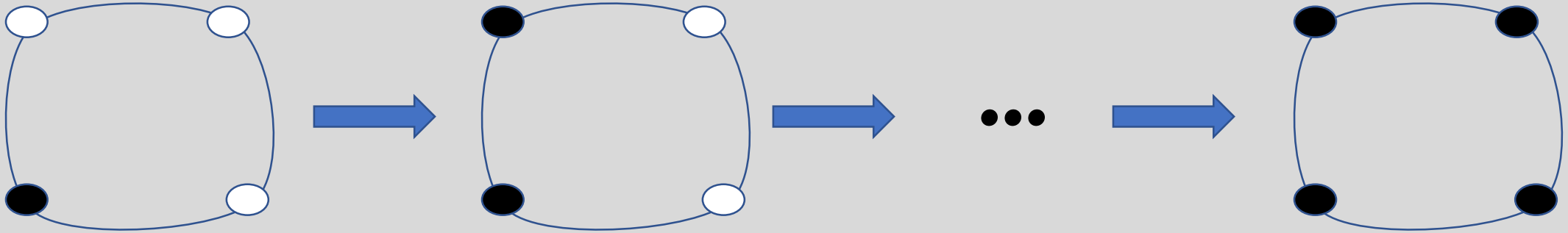
Cell in Graph



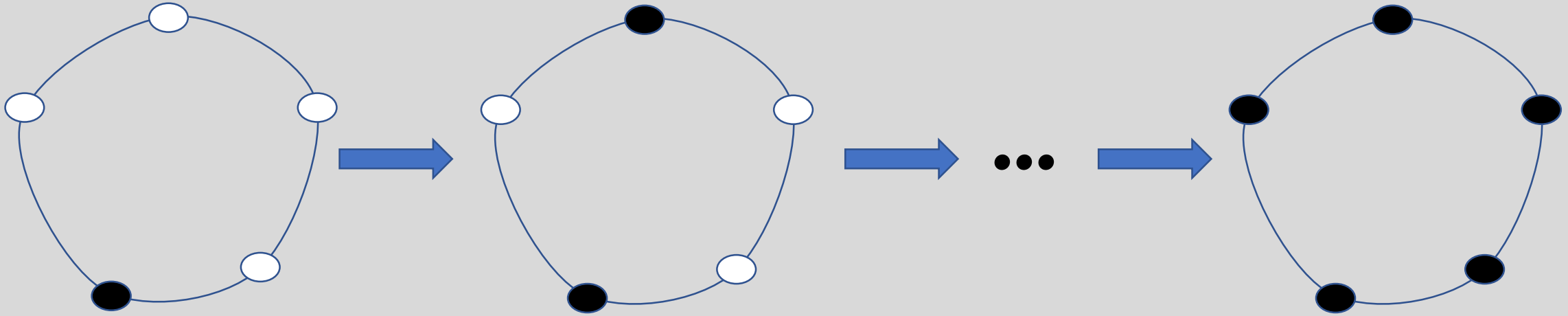


step	final
3	0
2	1

n=4



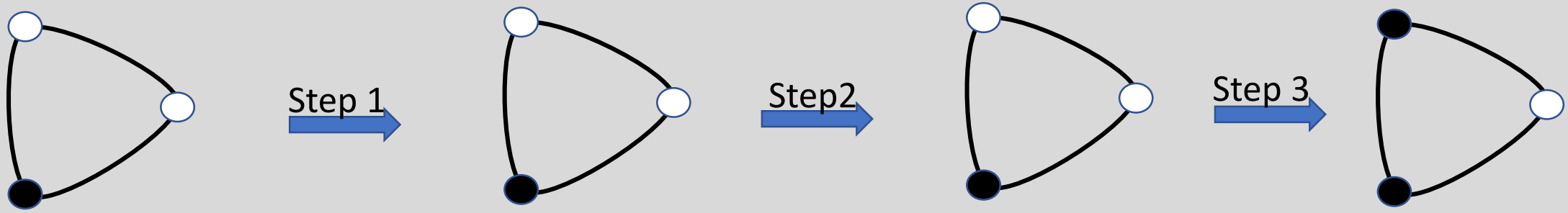
n=5



- n:Number of Graph Vertices

- ɣ Gamma: A concept of cell contagiousness (0.1,1)

ɣ =0.1

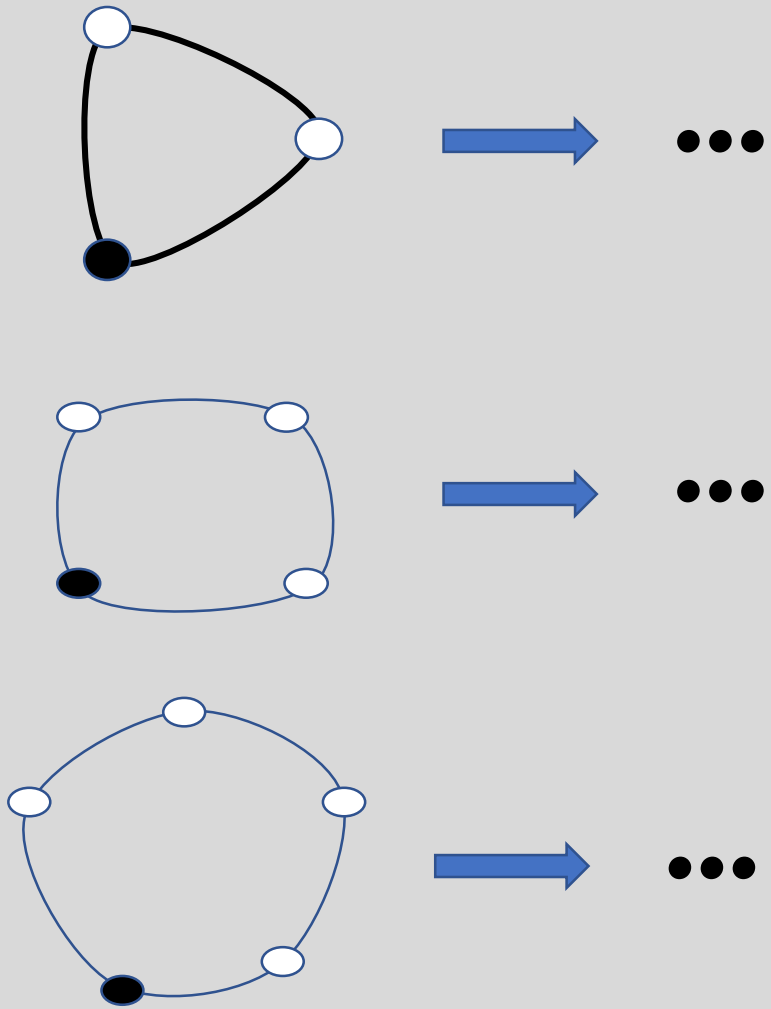


ɣ =0.9



Simulation

For Each Gamma and n



n=3 & ɣ =0.1

step	final
13	1
20	0
17	1
…	…

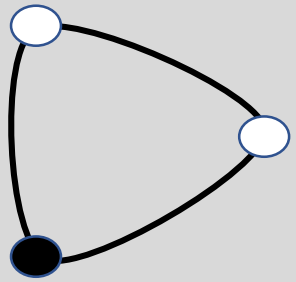
n=3 & ɣ =0.2

step	final
10	1
11	1
8	0
…	…

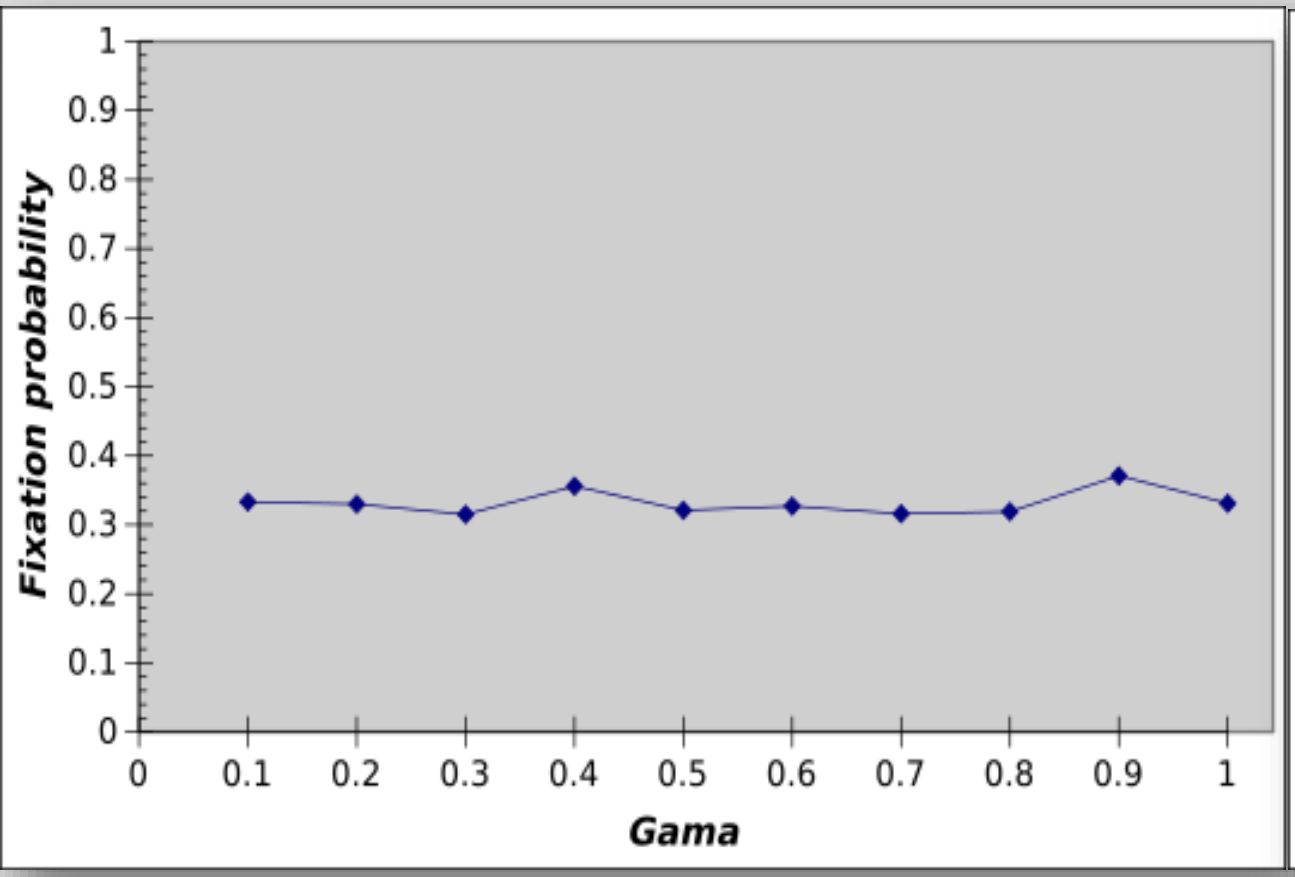
step	final
20	1
23	0
20	1
18	1
11	0
27	0
7	0
22	0
23	1
38	0

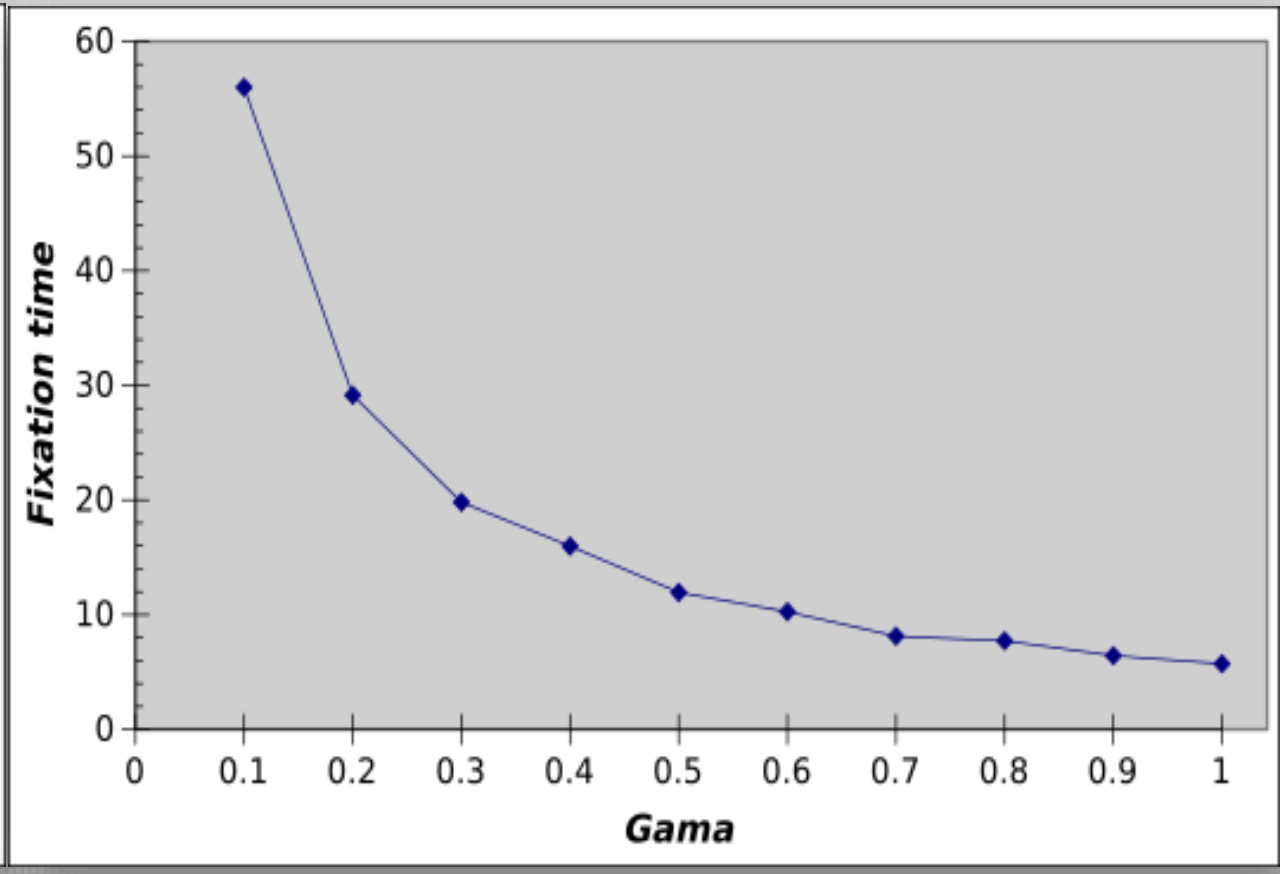
- **Fixation Probability :** $\dfrac{\text{The Number of Times the Graf becomes Fully 1}}{\text{Total Attenmpts}} = \frac{4}{10} = 0.4$

- **Fixation Time :** Average Steps of Fully Reached States $= \frac{20+20+18+23}{4} = 20.25$

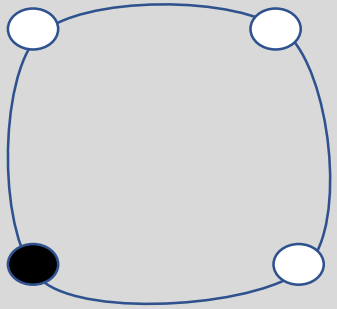


Simulation for n=3

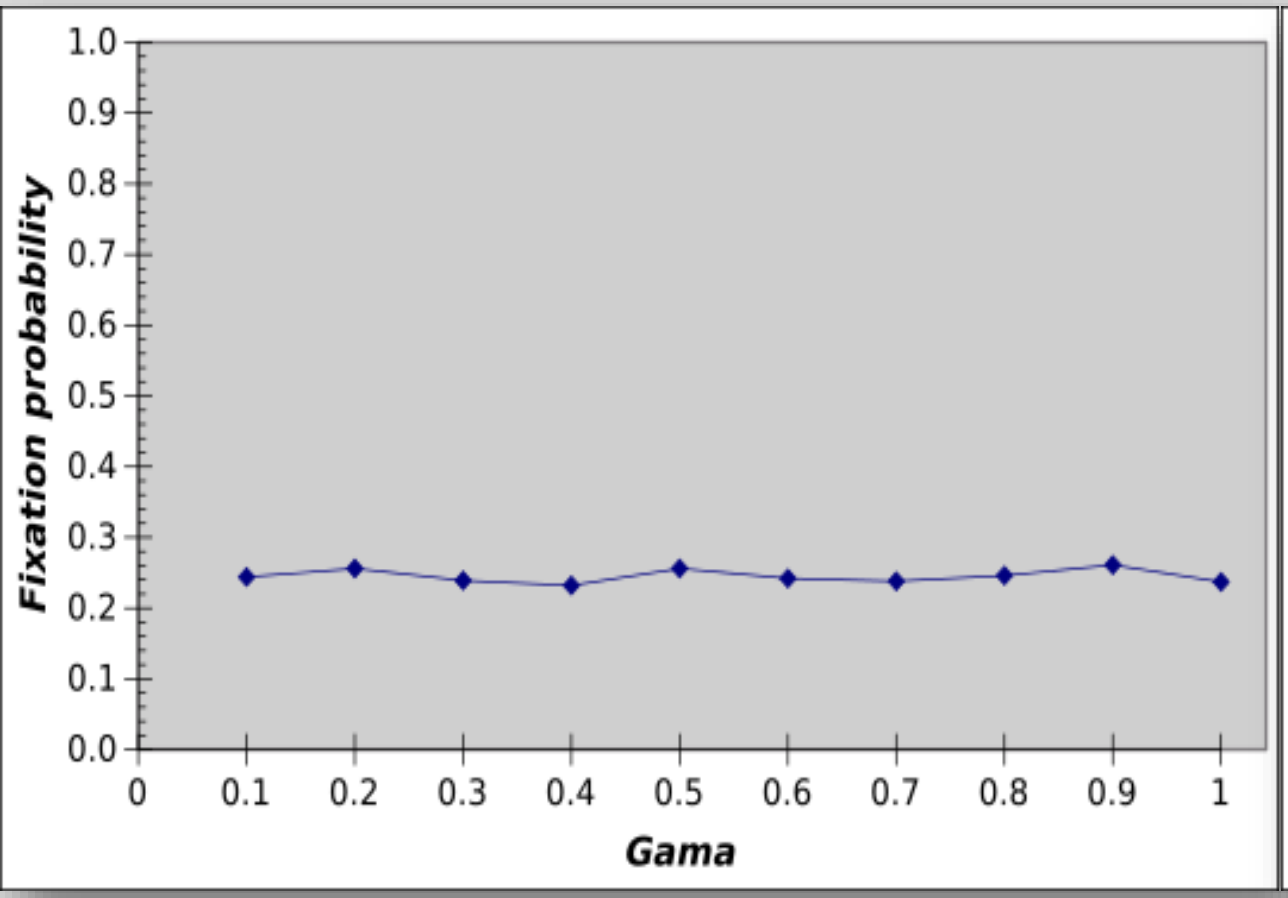


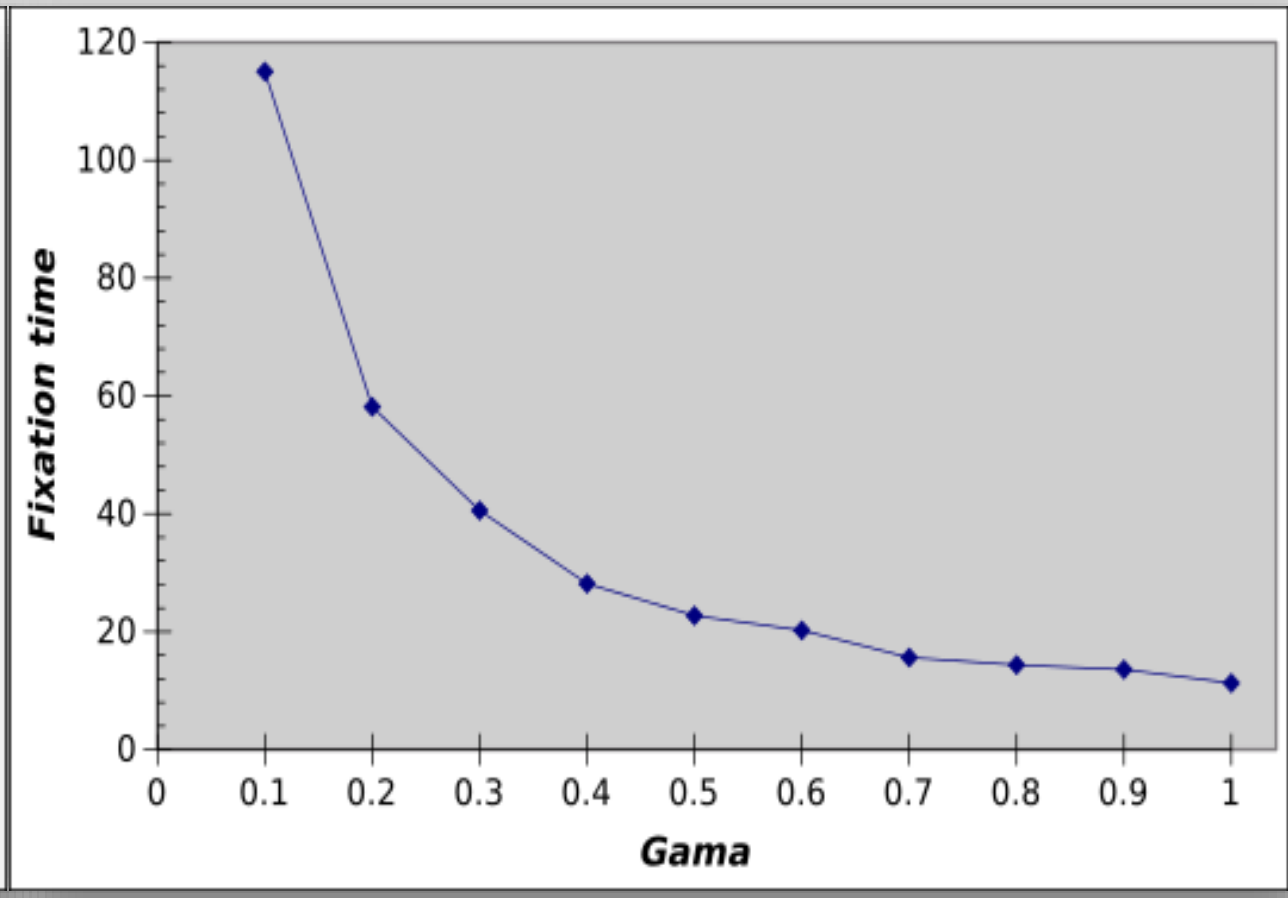


- Approximately Constant Around 0.3
- Decreasing with Increasing Gamma

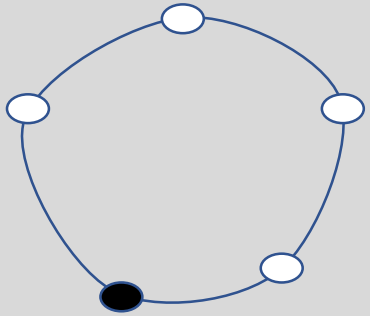


Simulation for n=4

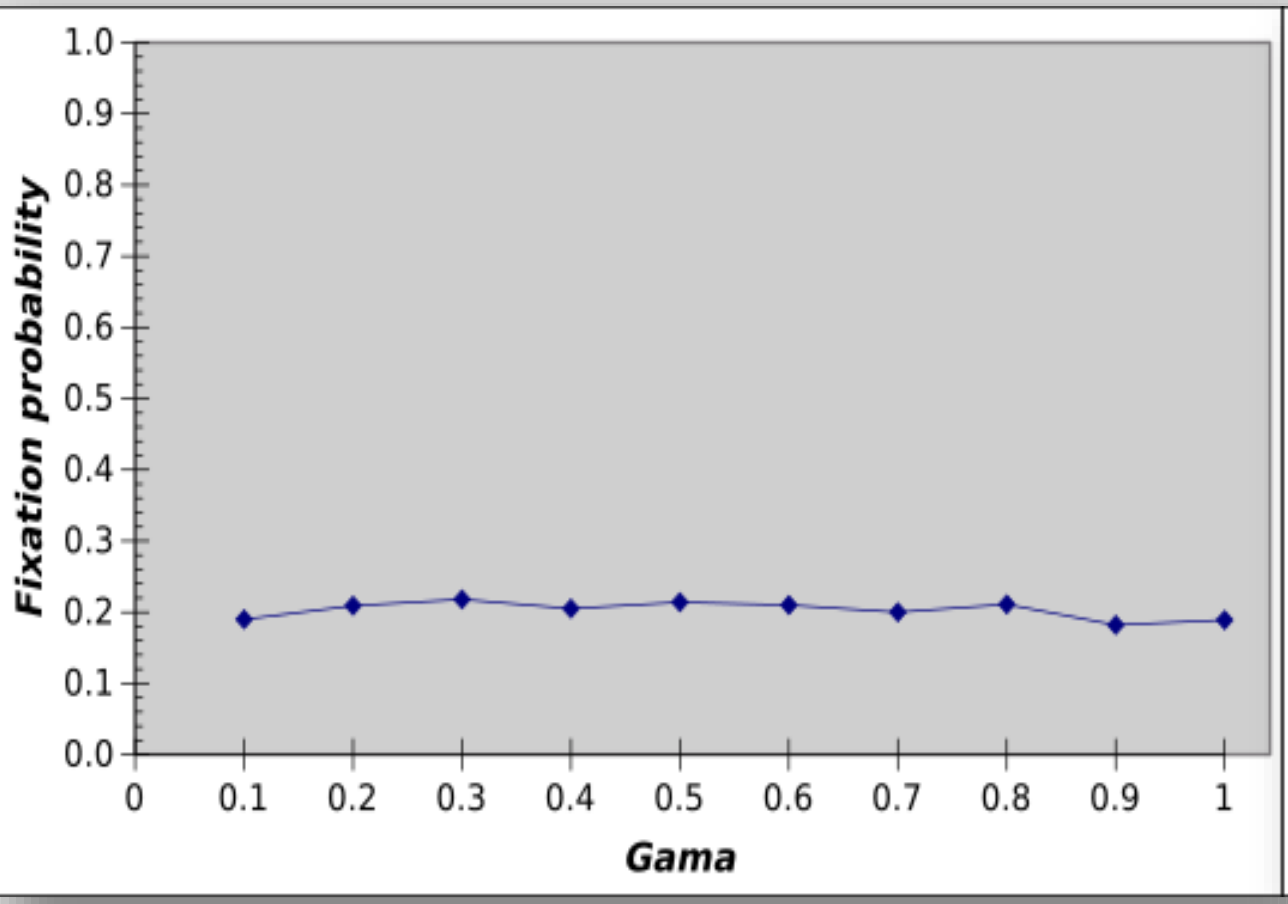


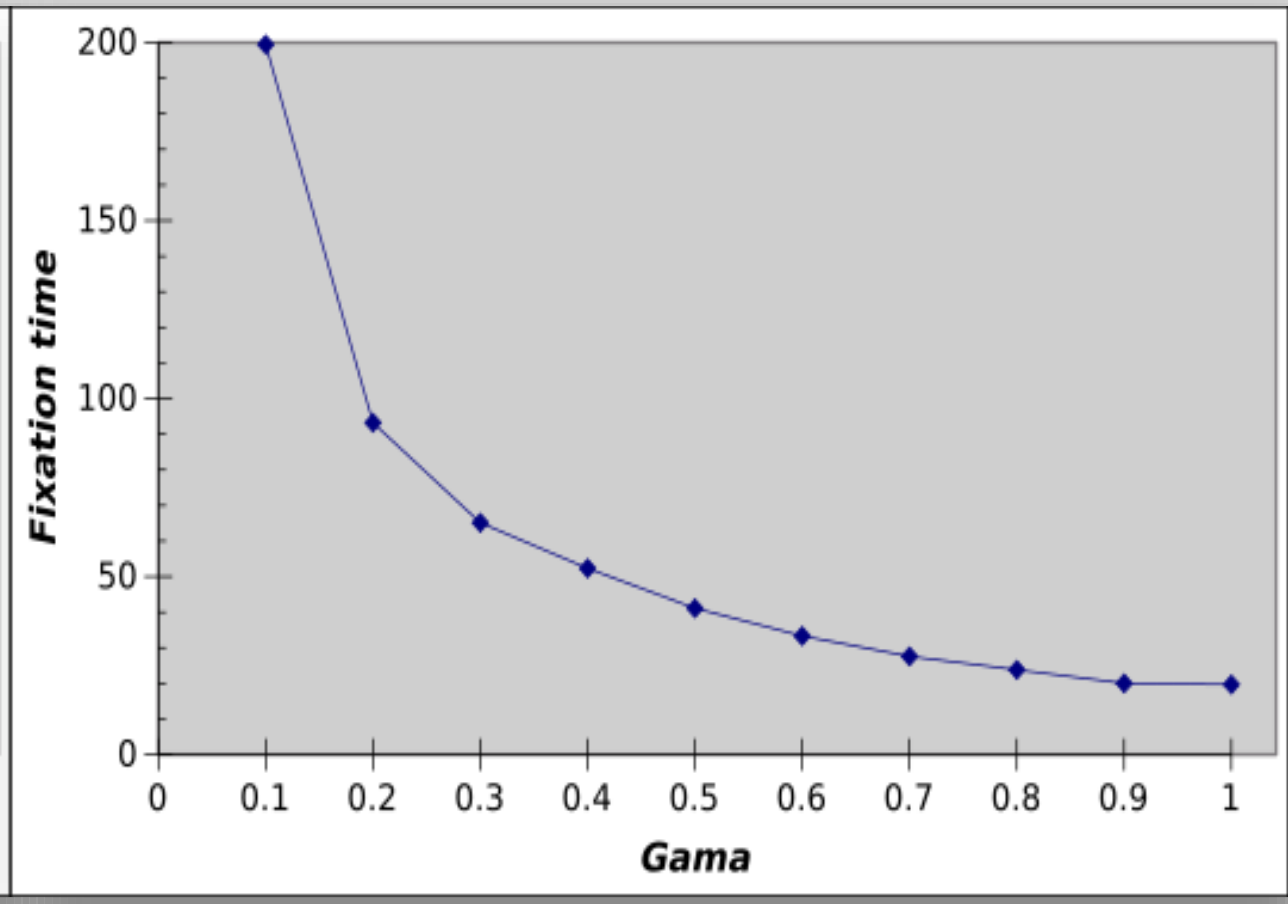


- Approximately Constant Around 0.25
- Decreasing with Increasing Gamma



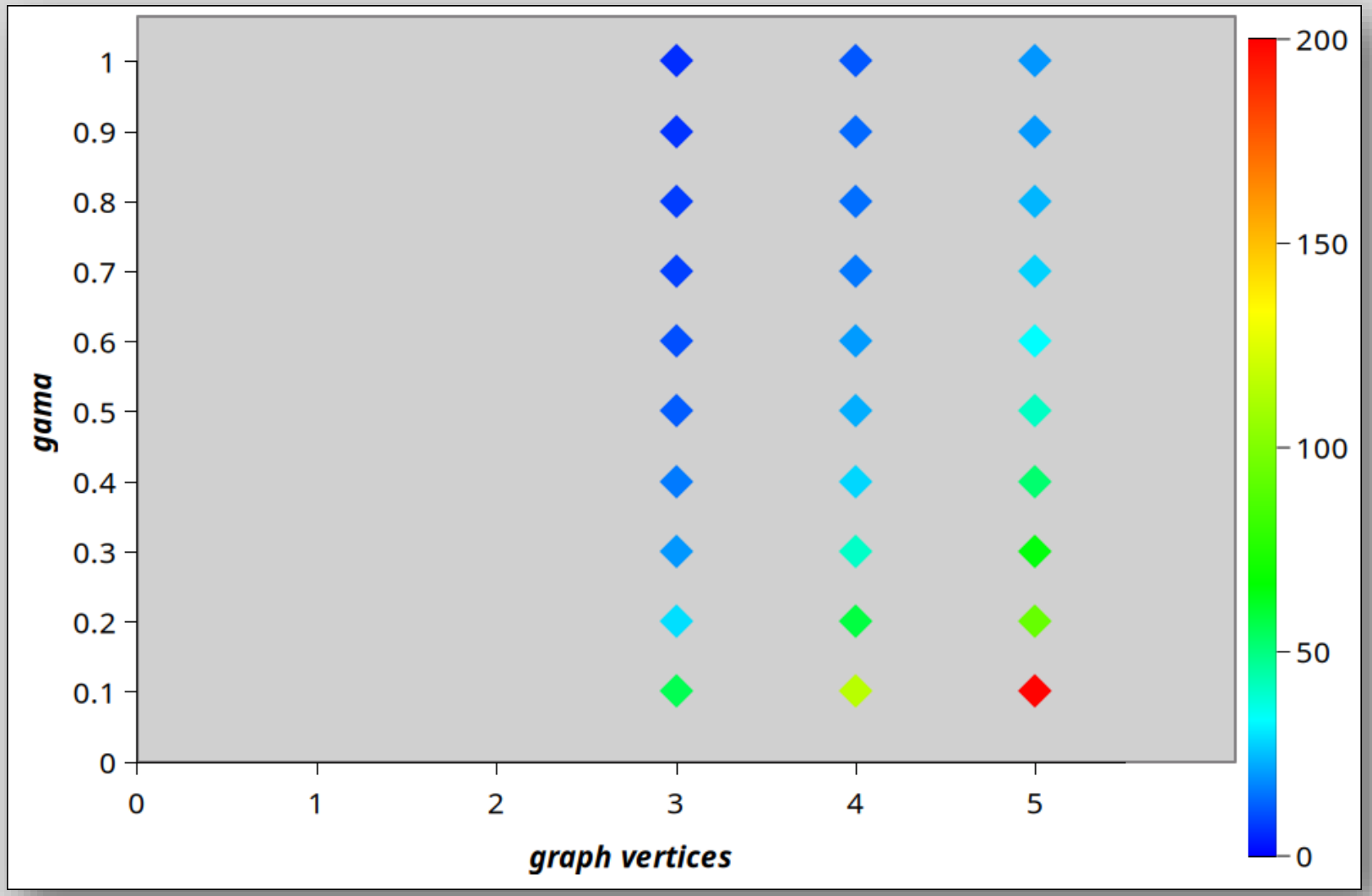
Simulation for n=5



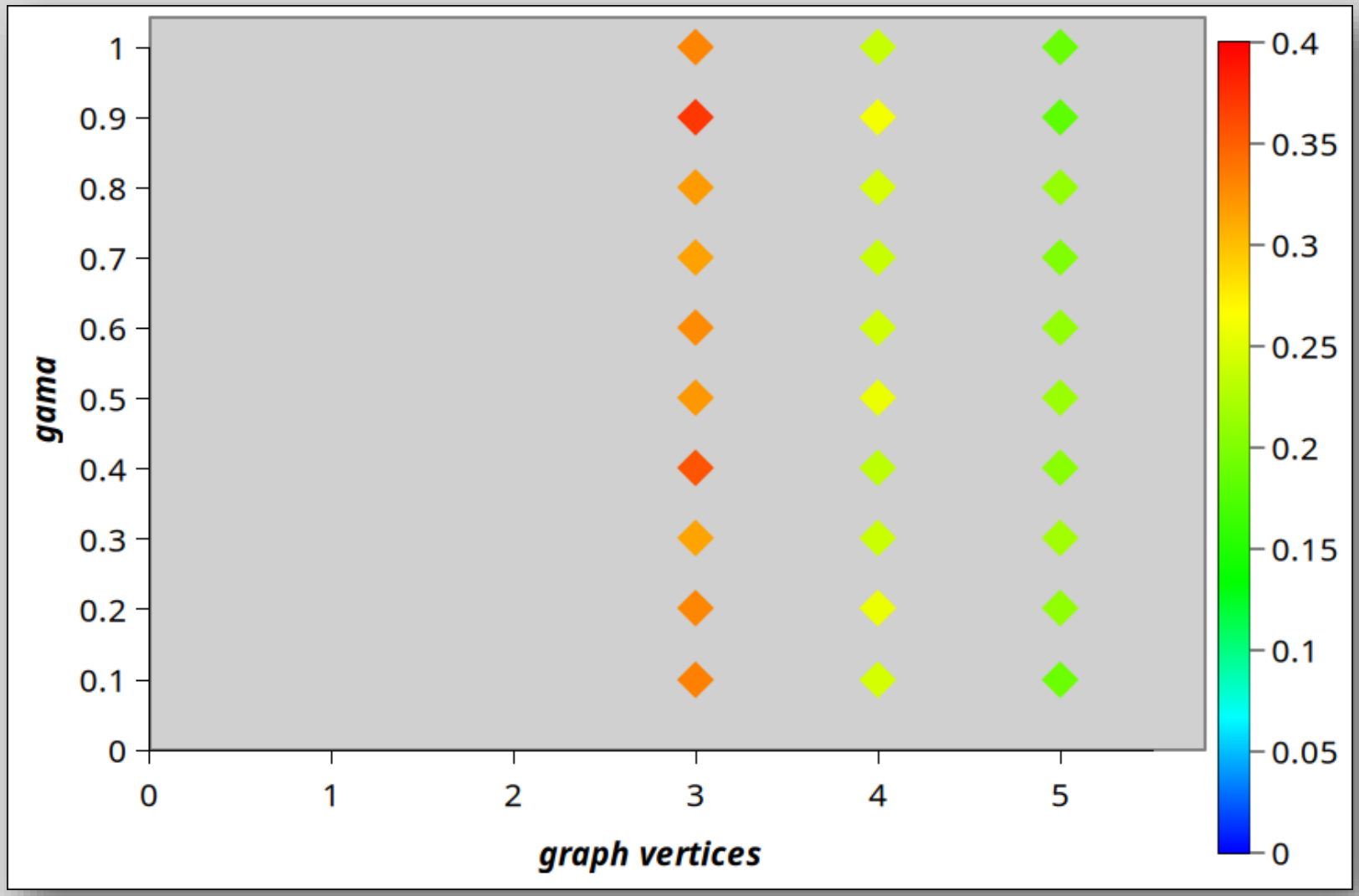


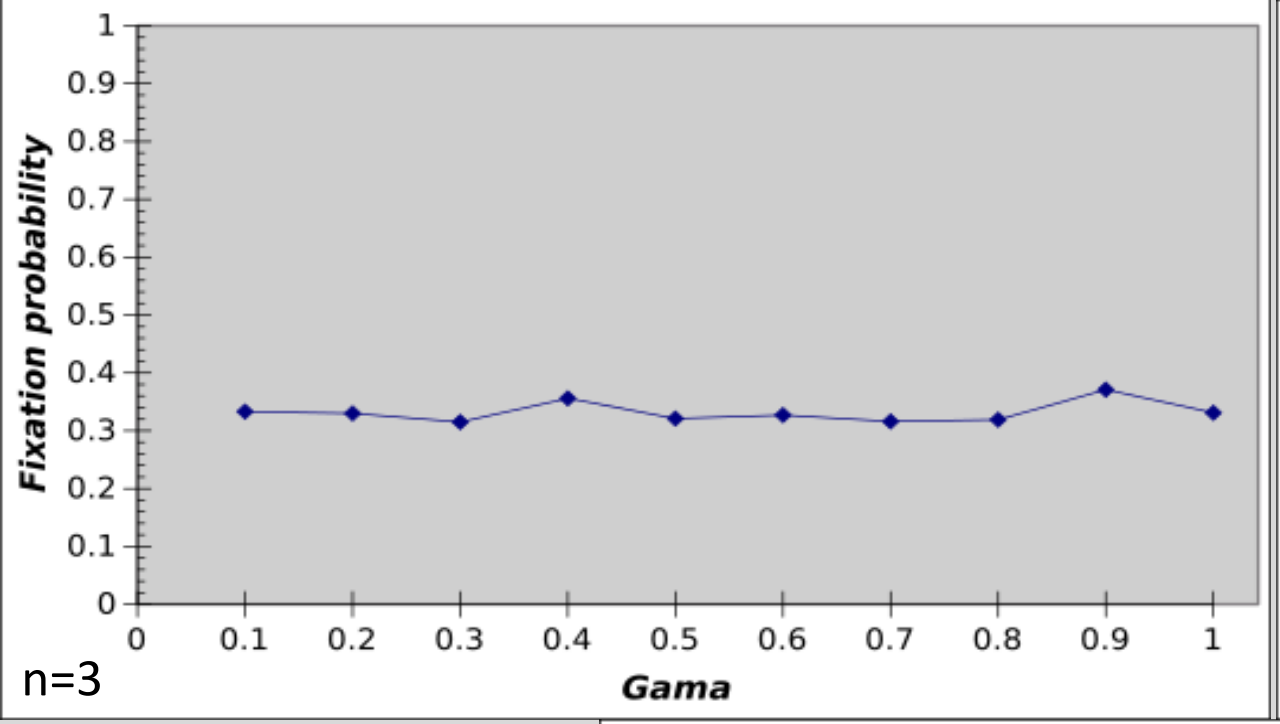
- Approximately Constant Around 0.2
- Decreasing with Increasing Gamma

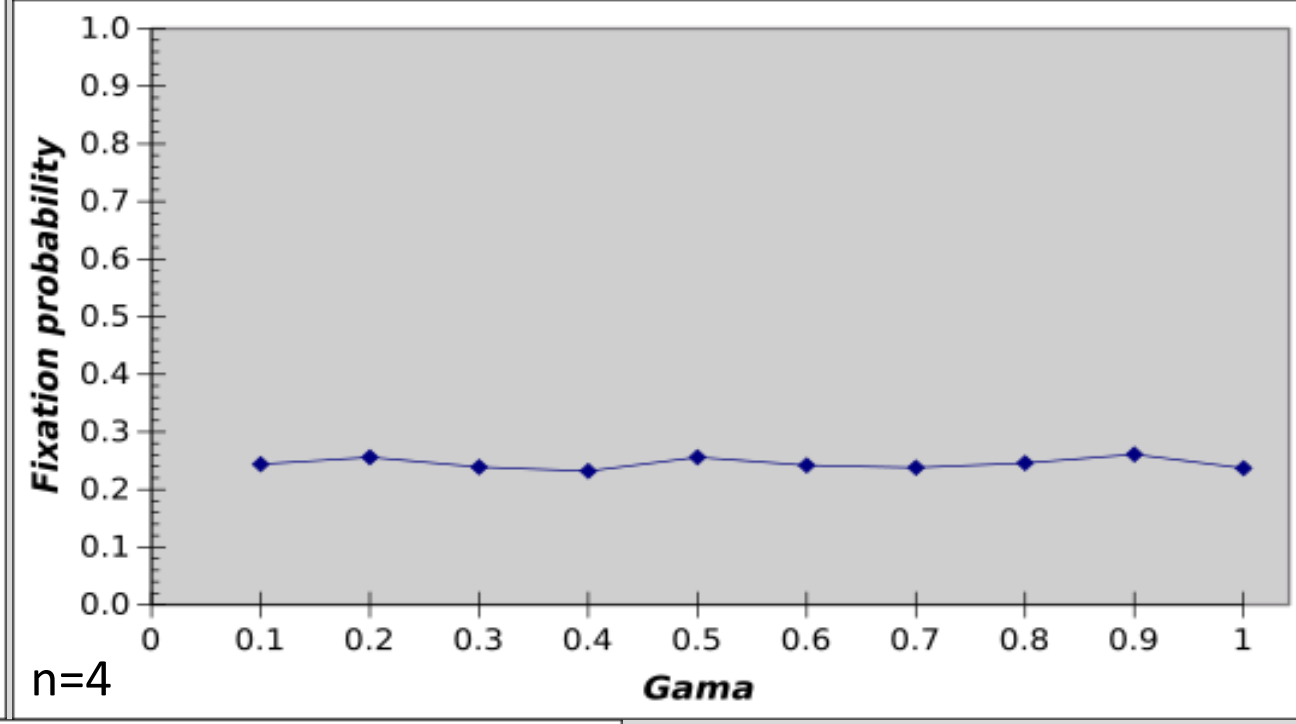
Fixation Time

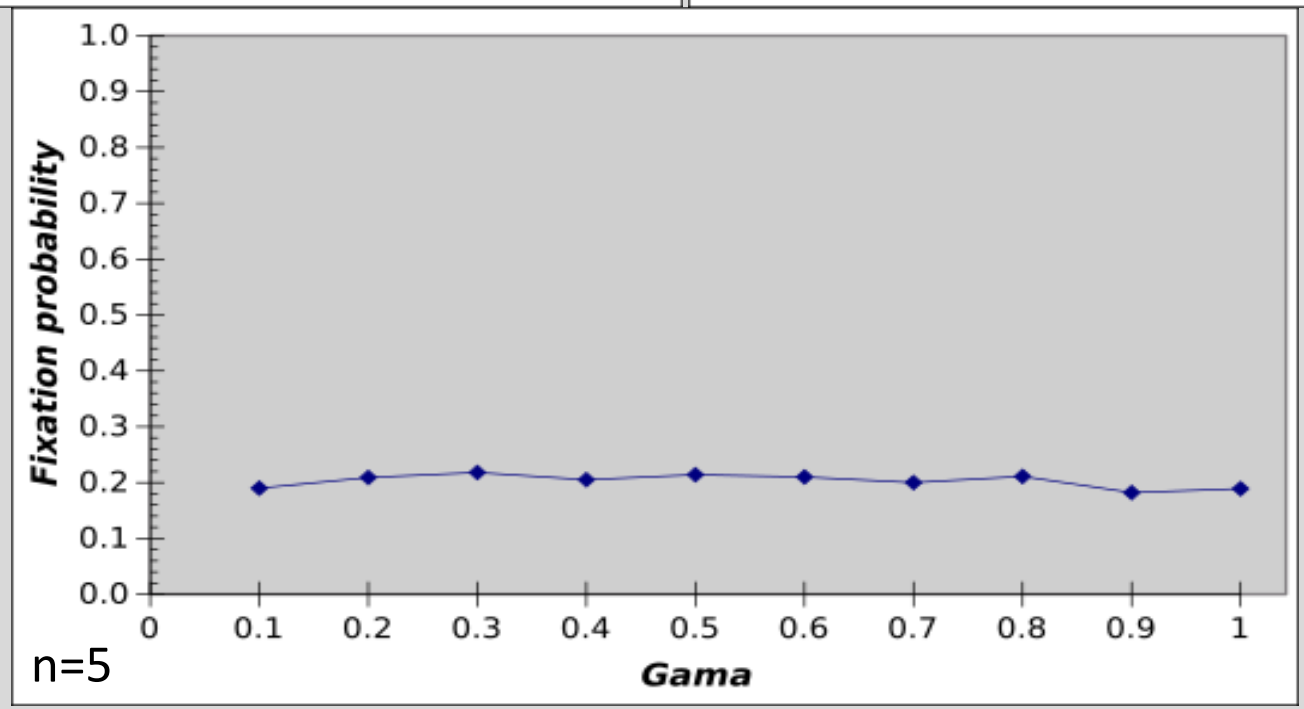


Fixation probability




Fixation probability
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Gama
n=3


Fixation probability
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Gama
n=4

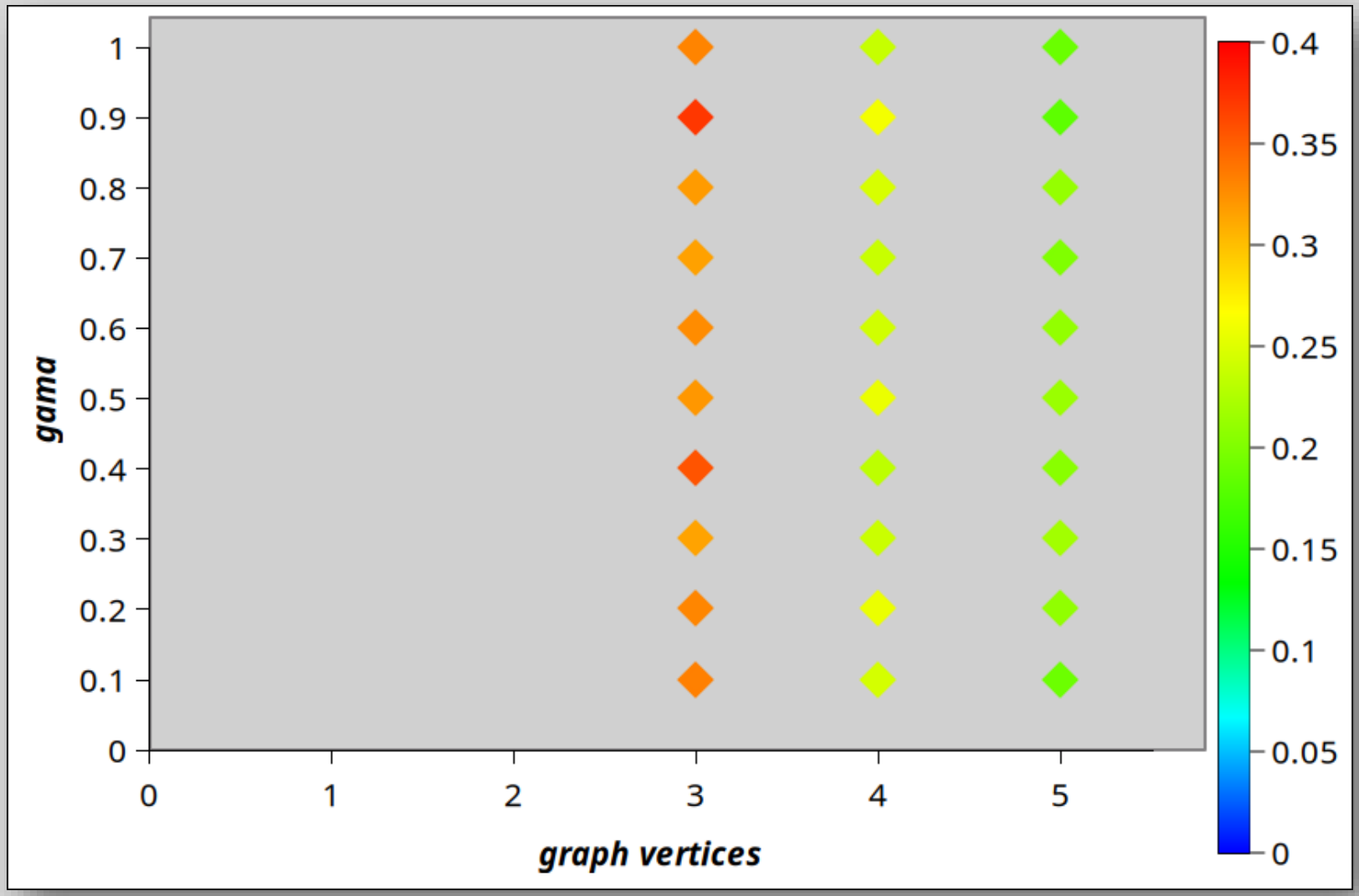

Fixation probability
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Gama
n=5

n	Fixation Probability
3	1/3 (0.3)
4	1/4 (0.25)
5	1/5 (0.2)

It Seems That the Fixation Probability is Obtained as Follows

$$\text{Fixation Probability} = \frac{1}{\text{graph vertices}}$$

Fixation probability



The End